# Vectors: Components and Operations

## Vector arithmetic

1 Let $\mathbf{u} = \langle 3, -2 \rangle$ and $\mathbf{v} = \langle -1, 4 \rangle$. Find:

a $\mathbf{u} + \mathbf{v}$  b $3\mathbf{u} - 2\mathbf{v}$  c $|\mathbf{u}|$

2 Find the component form and magnitude of the vector from $P(1, 3)$ to $Q(5, -2)$.

3 Find the unit vector in the direction of $\mathbf{v} = \langle 6, -8 \rangle$.

4 Write the vector of magnitude 12 making an angle of $150°$ with the positive $x$-axis in component form (exact values).

## Applications

5 A plane flies due east at 200 mph while the wind blows due north at 50 mph.

a Find the magnitude of the plane's resultant velocity, to one decimal place.

b Find the direction of the resultant, as an angle north of east to two decimal places.

6 Two forces $\mathbf{F}_1 = \langle 30, 0 \rangle$ N and $\mathbf{F}_2 = \langle 0, 40 \rangle$ N act on an object. Find the magnitude of the resultant force.

7 A boat aims to cross a river by heading due north at 4 m/s, but the current pushes it east at 1.5 m/s.

a Find the boat's resultant speed, to two decimal places.

b If the river is 200 m wide, find how long the crossing takes.

c Find how far downstream (east) the boat lands.

## Position vectors and geometry

8 Given $A(-2, 5)$ and $B(4, -3)$, find the midpoint of $\overline{AB}$ and the vector $\vec{AB}$.

9 Let $\mathbf{a} = \langle 2, 3 \rangle$ and $\mathbf{b} = \langle -4, 1 \rangle$. Find a vector $\mathbf{c}$ such that $\mathbf{a} + \mathbf{b} + \mathbf{c} = \langle 0, 0 \rangle$, and verify your answer.

10 Express $\mathbf{v} = \langle 5, -12 \rangle$ in terms of the standard unit vectors $\mathbf{i}$ and $\mathbf{j}$, and find $|\mathbf{v}|$.

## More applications

11 An airplane has an airspeed of 305 mph and flies on a heading $7.5°$ west of due north, while a 40 mph wind blows due east. Find the plane's resultant ground velocity vector, and its ground speed, to one decimal place (take east as the positive $x$-direction and north as the positive $y$-direction).

12 Given $\mathbf{u} = \langle 4, -1 \rangle$, find a vector orthogonal to $\mathbf{u}$ with the same magnitude as $\mathbf{u}$.

## Finding direction angles

13 Find the direction angle of $\mathbf{v} = \langle -3, 3\sqrt{3} \rangle$, to the nearest degree.

14 A vector has magnitude $10$ and direction angle $200°$. Write it in component form, rounded to two decimal places.

## Vector equations and unknowns

15 Find scalars $c_1$ and $c_2$ such that $c_1\langle 1, 0 \rangle + c_2\langle 0, 1 \rangle = \langle -3, 7 \rangle$.

16 Find scalars $c_1$ and $c_2$ such that $c_1\langle 2, 1 \rangle + c_2\langle 1, 3 \rangle = \langle 7, 11 \rangle$.

## Parallel and equal vectors

17 Determine whether $\mathbf{u} = \langle 4, -6 \rangle$ and $\mathbf{v} = \langle -6, 9 \rangle$ are parallel, and if so, state whether they point in the same or opposite directions.

18 Find the value of $k$ so that $\langle k, 6 \rangle$ is parallel to $\langle 4, 8 \rangle$.

## Vectors in three real-world scenarios

19 A hiker walks $3$ km east, then $4$ km north, then $2$ km west. Find the magnitude of the hiker's total displacement from the starting point.

20 Three ropes pull on a ring with forces $\langle 10, 0 \rangle$, $\langle -5, 8.66 \rangle$, and $\langle -5, -8.66 \rangle$ N. Find the net force, and explain what this implies physically.

21 A crane hoists a load using two cables. One cable exerts a force of $\langle -200, 350 \rangle$ N and the other exerts $\langle 200, 400 \rangle$ N. Find the combined lifting force (the vertical component of the resultant), and explain why the horizontal components matter for the crane's structural design even though they cancel.

**22** A swimmer aims to swim directly across a river at $1.5$ m/s relative to the water, while the current flows at $0.8$ m/s downstream. Find the swimmer's resultant speed relative to the shore, and the angle (from the intended straight-across direction) at which they actually drift, to the nearest degree.